

Phase 6: Execution Demonstration

1. Initial State and UI Generation

The EGL template successfully generates the user interface by traversing the Pivot nodes and edges. The game engine initializes correctly, reading the defined starting node, default boolean variables, and configuring an empty inventory set.

The screenshot displays a game interface titled "Frank's Dark Room" with the subtitle "Interactive Story Flow". The main content area is titled "FRANK NEGOTIATION" and contains the text: "Frank eyes you suspiciously across the hood of his RV. The key glints on his belt loop. How do you approach this?". Below this text is a section titled "WHAT DO YOU DO?" with two selectable options: "Negotiate peacefully" and "Threaten Frank 🗡️". At the bottom of the interface, there are two panels: "INVENTORY" showing "• Empty" and "FLAGS" showing "dark_room_evidence: false". A "Restart Game" button is located at the very bottom.

Figure 1: Initial game state at the 'Frank Negotiation' node.

2. Guard Evaluation (Locked Choices)

Choices with unmet conditions are rendered but explicitly disabled. This demonstrates that the abstract guard expressions are correctly translated into executable JavaScript logic that blocks narrative progression.

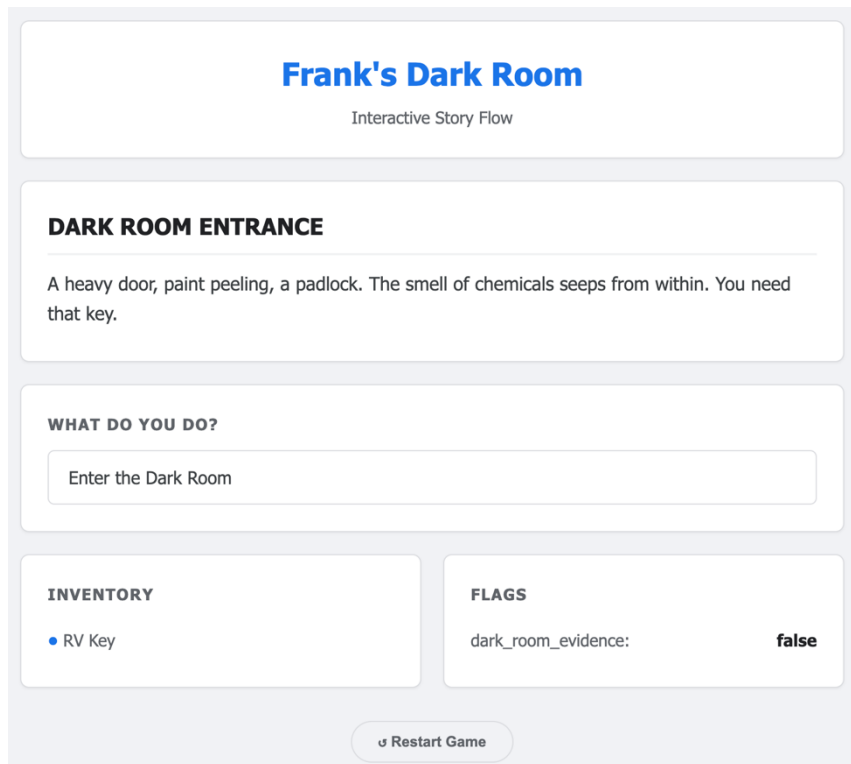


Figure 2: Guard condition successfully blocking unauthorized scene transition.

3. Dice Rule Execution & Effects

The probabilistic mechanics defined in the HLL and translated to the Pivot execute flawlessly. Upon selecting a choice with a DiceRule, the engine generates a random integer within the defined bounds. If the threshold is met, the associated actions are applied to the state.

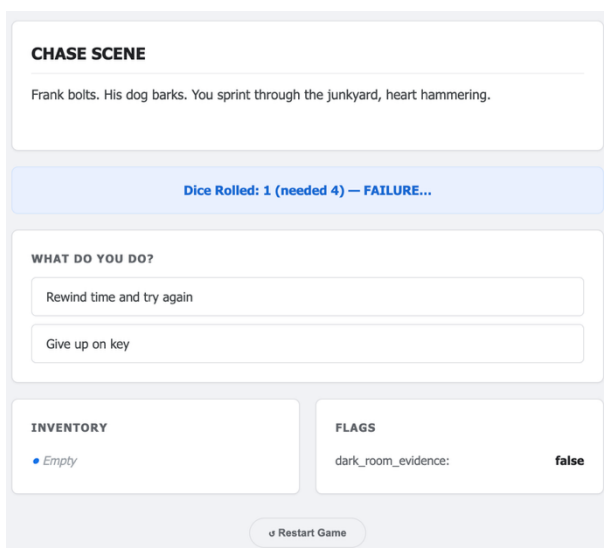


Figure 3: Dice roll failure – didn't get the key.

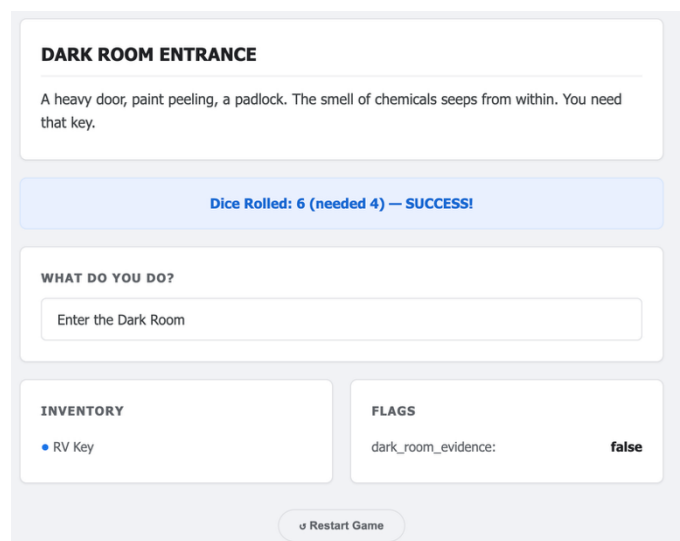


Figure 4: Dice roll success – key in the inventory.

4. State Persistence and Progression

With the required item obtained from the previous interaction, the engine re-evaluates the guard conditions. The previously locked choice dynamically unlocks, proving that state persistence and edge evaluation work in tandem.

INSIDE DARK ROOM

Red light. Rows of photographs. Evidence of every victim. Your stomach turns as you reach for the stack.

WHAT DO YOU DO?

Collect photo evidence

INVENTORY

- RV Key

FLAGS

dark_room_evidence: **false**

[↶ Restart Game](#)

Figure 5: Dynamic unlocking of choices based on satisfied state dependencies

LIGHTHOUSE PATH

The cliff path winds upward through salt grass. Below, waves crash against the rocks of Arcadia Bay.

WHAT DO YOU DO?

Walk to the lighthouse cliff

INVENTORY

- RV Key
- Photo evidence

FLAGS

dark_room_evidence: **true**

[↶ Restart Game](#)

Figure 6: Dynamic unlocking of choices based on satisfied state dependencies

5. Terminal Nodes

Reaching an accepting node correctly halts the narrative graph traversal. The interface hides further choices and displays the predefined terminal state badge (GOOD, BAD, or NEUTRAL), validating the structural integrity of the generated graph.

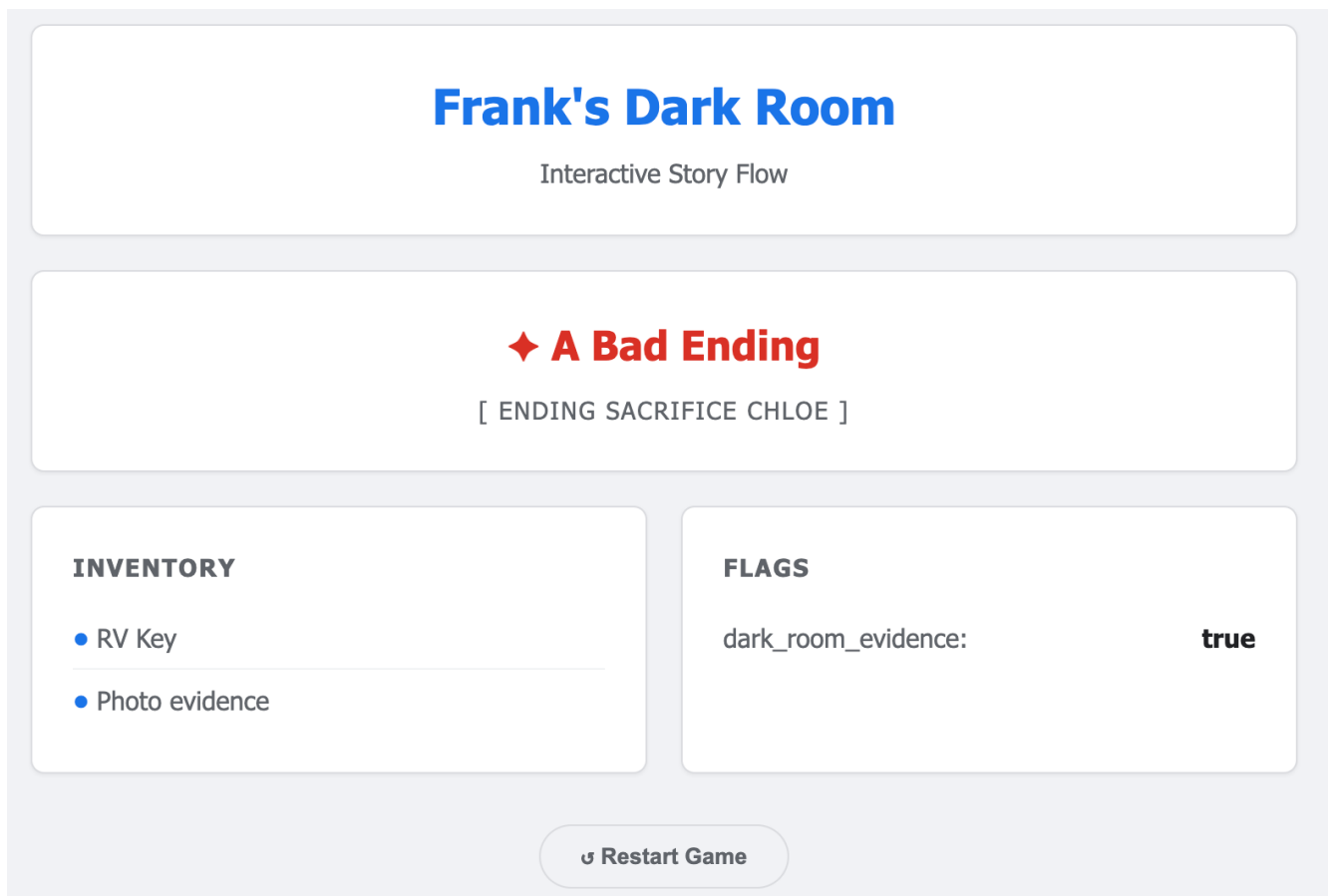


Figure 7: Terminal state execution showing the respective ending evaluation.